

Injection Moulding Quality Prediction

Automated Defect-Type Classification Using the ProBayes Dataset

A CRISP-DM Approach

Automating the Inspection Decision

Problem Statement:

Injection moulding produces plastic parts at high speed and volume. Defects occur when process parameters drift. Currently, a human inspector examines every part after ejection and assigns a defect label — subjective, operator-dependent, and not scalable with machine throughput.

What we solved:

Predict which specific defect type is present per cycle: not just OK or NOK: using sensor data already collected automatically on the production line.

Current Process

Inject & hold

Part ejected

IR camera scan

Scale (weight)

Vision camera

Human inspector ← bottleneck

Defect label assigned

Subjective · slow · not scalable



What We Built — Triage Layer

Sensor data collected automatically

MODEL runs

All OK

Part passes

**High-conf
defect**

Auto-flag

Borderline

Human review
(reduced load)

Human review reduced to borderline cases only

Dataset: ProBayes | SKZ + Fraunhofer IPA, 2021–2022

564

Parts (cycles)

47

Experimental conditions

2

Materials (PP + ABS)

334

Raw features

9

Sensor sources

Design of Experiment — Four Parameters Varied

Cylinder temperature · Mould temperature
Injection speed · Holding pressure

Challenge: Experiment Group Structure

MET_ExperimentNumber repeats across materials

"A01" = PP December 2021 ≠ "A01" = ABS December 2021

Using number alone: 30 groups

Correct count: 47 unique conditions

Fix: EXP_GROUP = ExperimentNumber + Material + Date

Wrong key → evaluation leakage before any model trains

What We Predict: Six Specific Defect Types

Defect	Prevalence	Physical Cause
SinkMarks	25.4% (143/564)	Inadequate packing: surface collapse during cooling
SprueCircle	12.8% (72/564)	Overpacking at gate: stress concentration at sprue
Underfilled	10.6% (60/564)	Insufficient cavity fill: material did not reach all areas
StreaksLevel1	5.3% (30/564)	Flow lines: sluggish material flow, cold/slow process
StreaksLevel2	3.2% (18/564)	Pressure instability: oscillating injection pressure
StreaksLevel3	8.5% (48/564)	Burn marks: material overheating and carbonisation
NOK (derived)	27.5% (155/564)	SinkMarks OR Underfilled: per dataset definition, no separate model

OldGranulate excluded — contamination event, not a process defect (27% values missing).

Overall Pipeline

SENSOR INPUTS · collected automatically on the production line

E77 (cycle scalars) · SET (DoE parameters) · ENV (ambient) · IR (thermal) · CV (geometry) · SCA (weight) · DXP (pressure curves)



FEATURE ENGINEERING

334 raw features → 63 scalar features + DXP time series (2 channels × 5,331 timesteps)



LightGBM

63 scalar features
SinkMarks · SprueCircle · Streaks L1/2/3



TCN

Raw DXP curves · 5,331 timesteps × 2 channels
Underfilled



6 DEFECT PREDICTIONS per cycle · SinkMarks · SprueCircle · Underfilled · StreaksL1 · StreaksL2 · StreaksL3 · + NOK derived

Evaluated under GroupKFold(5) by experimental condition throughout

Feature Engineering

Source	Features	What it measures
E77	23	Cycle-level: pressures, temperatures, times
SET	4	DoE parameters: 4 varied settings
ENV	2	Ambient: air temperature, humidity
IR	23	Part surface temperatures (Images 1 + 2)
CV	10	Geometric dimensions: heights, widths, warpage
SCA	1	Part weight
MAT	1	Material type (PP / ABS)

334→63: structural missingness dropped · near-zero variance dropped · high correlation dropped

DXP Pressure Curves — Diagnostic Finding

Raw cavity pressure: 4 phases per cycle

Injection spike → Holding plateau → Gate freeze → Cooling decay

35 phase-based features extracted via trigger masks.
Then tested across 3 runs:

Feature set	SinkMarks	Underfilled
Scalar only	0.915 ✓	0.539
Combined	0.835	0.620
DXP only	0.779	0.704 ✓

*DXP scalars redundant with E77 for most targets.
Exception: Underfilled — curve shape irreplaceable. → TCN.*

Model Selection: Two Models for Different Reasons

Target	Model	Why
SinkMarks, SprueCircle, Streaks	LightGBM (scalar features)	E77 scalars capture the mechanism: scalar model sufficient
Underfilled	TCN (raw DXP curves)	Curve shape during filling is irreplaceable: +0.203 F1 over scalar

LightGBM on Scalar Features

Sequential error correction

Each tree corrects residual errors of the previous one.

On 564 samples with prevalence as low as 3.2%, boosting extracts minority class signal that simple averaging (Random Forest) misses.

TCN on Raw DXP Curves

Dilated causal convolutions

4 blocks (dilation 1, 2, 4, 8) — patterns at multiple timescales.

Reads the full waveform without hand-crafted choices.
146,983 parameters · early stopping · 5,331 timesteps × 2 channels.

Underfilled: F1 = 0.742 vs scalar F1 = 0.539 (+0.203)

Evaluation Strategy: Measuring Generalisation

Random 70/30 Split

Parts from the same experimental condition appear in both train and test.

Model recognises group patterns — not process physics.

Result: near-perfect scores.

GroupKFold(5) by EXP_GROUP

Each test fold contains ~9 experimental conditions the model never saw in training.

Tests cross-condition generalisation.

Honest estimate for production deployment.

The gap between them IS the leakage:

Defect	Random Split	GroupKFold	Gap
SinkMarks	1.000	0.915	-0.085
SprueCircle	1.000	0.848	-0.152
Underfilled	1.000	0.742	-0.258

We started with random split — near-perfect scores. Then identified why. GroupKFold is the primary evaluation throughout.

Results: GroupKFold(5) Cross-Condition Generalisation

Defect	Model	F1 mean	F1 std	Key driver
SinkMarks	LightGBM scalar	0.915	± 0.060	E77_CushionVolume
SprueCircle	LightGBM scalar	0.848	± 0.219	SCA_PartWeight + IR dome temp
Underfilled	TCN raw DXP	0.742	± 0.246	Pressure curve shape
StreaksLevel1	LightGBM binary	0.827	± 0.213	E77_TransferPressure
StreaksLevel2	LightGBM binary	0.786	± 0.262	E77_InjectionPressureMax
StreaksLevel3	LightGBM binary	0.887	± 0.225	IR_Img2TempVerRng
NOK (derived)	SinkMarks OR Underfilled	0.797	± 0.050	—

SinkMarks: strongest and most consistent (std ± 0.060). **NOK derived:** most stable — OR combination smooths individual fold variance (std ± 0.050).

StreaksLevel2: 18 positives only — treat as indicative. High fold variance = structural dataset constraint, not model failure.

Key Discovery: Streaks Are Not an Ordinal Severity Scale

Initial Assumption

Streaks = severity scale
Level 1 mild → Level 2 moderate → Level 3 severe
Single 4-class ordinal model.

What Data Told

Level 3: 48 positives
Level 1: 30 positives
Level 2: 18 positives
Reversed distribution for a severity scale.

Level	Mechanism	Top Feature	Direction
Level 1	Flow lines	E77_TransferPressure	Low pressure ↑ risk (cold/slow)
Level 2	Pressure instability	E77_InjectionPressureMax	High peak + low total ↑ risk
Level 3	Burn marks	IR_Img2TempVerRng	High temp variation ↑ risk (overheating)

Level 1 and Level 3 caused by opposite conditions: ordinal model receives contradictory gradient signals.

	Ordinal (before)	Binary (after)	Change
StreaksLevel1	0.743	0.827	+0.084
StreaksLevel2	0.743	0.786	+0.043
StreaksLevel3	0.743	0.887	+0.144 ← largest gain

Discovery came from data. Corrected mid-project. Verified by improved results.

SHAP: What Drives Each Defect Prediction

SinkMarks

E77_CushionVolume

Screw bottoms out → no holding pressure reaches cavity → material shrinks → surface depression. IR temperature gradient features also appear — IR camera detects uneven cooling non-uniformity.

Underfilled

E77_AreaMeltPressure + E77_TransferStroke

Total pressure energy + screw fill distance — together capturing whether the cavity was reached. Actual machine behaviour more predictive than intended settings.

SprueCircle

SCA_PartWeight + IR_Img2TempDomeDiff

Heavier part = more material packed = more gate stress → circular mark at sprue. Two independent sensors (scale + IR) both detecting the overpacking signature.

StreaksLevel3

IR_Img2TempVerRng

Vertical temperature variation across part surface — direct thermal signature of localised overheating and carbonisation detected by IR camera.

Every prediction traceable to a physical process variable — actionable for process engineers.

Two Further Findings

Finding 1 — Different Defects Need Different Sensors

Defect	Critical Sensor	Why
SinkMarks	E77 + IR temp gradients	IR detects uneven cooling non-uniformity at defect site
SprueCircle	SCA weight + IR dome temp	Two independent sensors both detecting overpacking
Underfilled	DXP raw curves (TCN)	Scalar summaries miss filling dynamics — curve shape essential
StreaksLevel3	IR thermal variation	Overheating signature detectable post-ejection

Excluding any sensor source degrades at least one model.

Finding 2 — Material-Specific Models Do Not Help (counter-intuitive)

Hypothesis: PP and ABS have different rheology — separate models should outperform the mixed model.

Test: trained PP-only and ABS-only models. **Mixed model wins or ties on every evaluable target.**

Why: defects are concentrated by material (all SprueCircle/Underfilled in PP; all Streaks in ABS). Splitting removes cross-material signal.

LightGBM already learns material-conditional patterns through tree splits on the binary material feature.

Conclusion: *material-specific modelling viable at 500+ cycles per material.*

Limitations and Future Work

Limitations

Data volume

564 parts / 47 conditions — high fold variance on rare defects.
Tuning confirmed num_leaves=15 optimal across all targets — shallow trees required, confirming dataset size as the binding constraint.

StreaksLevel2

18 positives — results indicative, not conclusive.

Single machine and geometry

Retraining required for any new production setup.

No drift detection

Model validity under process drift not evaluated.

Future Work

More production cycles

Stabilise variance · enable TCN across all targets · unlock material-specific modelling (500+ cycles per material).

Per-defect threshold tuning

Optimise precision/recall balance based on cost of false reject vs false pass.

Drift detection

Flag when input distributions shift beyond training range — early warning before model degrades.

Cross-machine generalisation

Test transfer across different KraussMaffei configurations.

Conclusion

01

What we built

Six defect-specific classifiers automating the injection moulding inspection decision — reducing human review to borderline cases only, with every prediction physically interpretable through SHAP.

02

What we achieved

F1-macro 0.742 to 0.915 under honest cross-condition evaluation (GroupKFold). The gap between random split ($F1 \approx 1.0$) and GroupKFold results quantifies what honest evaluation costs — and why it matters.

03

What we discovered

Streaks severity levels are three distinct defect mechanisms — flow lines, pressure instability, burn marks — not a progressive scale. Identified from SHAP mid-project, corrected, and confirmed by improved performance. Different streak types require different process adjustments.

Thank You!